Optimization HW-4

(1.1) Downward Spiral

P_T for one MALA step at temperature $T > 0$.
All measurable sets are subsets of K.

For arbitrary probability measures, including point-mass initialization use $\|\mu - \nu\|_{TV} = \sup_A |\mu(A) - \nu(A)|$

(1) Include rejection probability
Let

$$\gamma_T(x) = 1 - \int_K k_T(x,y)\,dy$$

Integral is probability of proposing & accepting a point in K.
All remaining outcomes stay at x

$$P_T(x,A) = \int_A k_T(x,y)\,dy + \gamma_T(x)\,1_A(x)$$

Includes proposals outside K and proposals rejected by the acceptance test.

(2) Stationarity of the Gibbs distribution.
Stationarity equation is

$$\pi_T P_T = \pi_T, \quad \int_K P_T(x,A)\,\pi_T(dx) = \pi_T(A) \quad \text{for every } A$$

Accepted density × Gibbs density

$$p_T(x)\cdot k_T(x,y) = \min\{p_T(x)\,q_T(x,y),\ p_T(y)\,q_T(y,x)\}$$
$$= p_T(y)\,k_T(y,x)$$

Thus, accepted moves satisfy detailed balance

Integrating in x, $\int_K p_T(x)\, k_T(x,y)\, dx = p_T(y)\cdot(1-r_T(y))$

The stayput term contributes $p_T(y)\, r_T(y)$, so sum is $p_T(y)$

(3) Gibbs mass concentrates near minimum

Positive Volume: Compactness & continuity give minimizer $x^* \in K$. continuity gives a neighborhood of x^* in which $f < f^* + \epsilon/2$.

Take a ball $B(z,r) \subset K$ which exists because K has non-empty interior. For sufficiently small $s > 0$, convexity gives

$$(1-s)x^* + sB(z,r) \subset K \cap \{f < f^* + \epsilon/2\} = A_\epsilon$$

The ball on left has radius $sr > 0$

Hence, $v_\epsilon = \mathrm{Vol}_d(A_\epsilon) > 0$, even if x^* is on the boundary

Probability bound: Bound numerator on B_ϵ and denominator on A_ϵ

$$\int_{B_\epsilon} e^{-f(x)/T}\, dx \le V_\epsilon e^{-(f^*+\epsilon)/T}, \quad Z_T \ge v_\epsilon e^{-(f^*+\epsilon/2)/T}$$

Dividing, $\Pi_T(B_\epsilon) \le \frac{V}{v_\epsilon} e^{-\epsilon/2T}$

④ A uniform lower bound on transition probabilities

Set $M = (D + hG)^2 / 4h$ so $c_M = \Delta + M$

For any $x, y \in K$, $\| y - x + h \nabla f(x) \| \leq D + hG$

same bound holds after exchanging x & y

$\therefore$ Both proposal densities are at least $4\pi h T^{-d/2} e^{-M/T}$

Using acceptance rule,

$$k_T(x,y) = \min \left\{ q_T(x,y), e^{-(f(y) - f(x))/T} q_T(y,x) \right\}$$

Since $f(y) - f(x) \leq \Delta$, both expressions in minimum are at least $(4\pi h T)^{-d/2} e^{-(M+\Delta)/T}$ Thus,

$$k_T(x,y) \geq \frac{\alpha_T}{V}, \quad P_T(x,A) \geq \alpha_T U_K(A)$$

Particularly, $0 < \alpha_T \leq 1$ because $P_T(x,K) = 1$

⑤ Total variation contracts at fixed temp.

The minorization allows the decomposition.

$$P_T(x,\cdot) = \alpha_T U_K(\cdot) + (1 - \alpha_T) R_T(x,\cdot)$$

where R_T is a transition kernel if $\alpha_T < 1$. The common uniform term cancels and transition kernels do not increase total variation.

$$\| \mu P_T - \nu P_T \|_{TV} = (1-\alpha_T) \| \mu R_T - \nu R_T \|_{TV}$$

$$\leq (1-\alpha_T) \| \mu - \nu \|_{TV}$$

If $\alpha_T = 1$, the kernel equals Π_x for every starting pt. and same bound holds with 0 on the right

(6) Consecutive Gibbs distribution changes slowly,

Let $\beta = 1/T$ $\tilde{\rho}_\beta$ for normalized density. Derivative (hint)

$$\int_K | \partial_\beta \tilde{\rho}_\beta(x) | dx = \mathbb{E}_\beta | f - \mathbb{E}_\beta f | \leq \Delta$$

because both f and its mean lie in $[f^*, f^* + \Delta]$.
Integrating derivative b/w two inverse temps yields

$$\| \pi_{1/\beta'} - \pi_{1/\beta} \|_{TV} \leq \frac{1}{2} \Delta | \beta' - \beta |$$

Here, $\beta_n = \log(n+2)/c$ Hence,

$$\delta_n \leq \frac{\Delta}{2c} \log \frac{n+2}{n+1} \leq \frac{\Delta}{2c(n+1)}$$

We use $\log(1+u) \leq u$

(7) contraction amounts have divergent sum,

Put $\rho = C_M/C < 1$ and $A = V(4\pi hc)^{-d/2} > 0$

By direct substitution $\alpha_n = A\left[\log(n+2)\right]^{d/2}(n+2)^{-\rho}$

For $n \geq 1$, $\log(n+2) > 1$ so, $\alpha_n \geq A(n+2)^{-\rho}$

The p series diverges for $\rho < 1$ and

$$\therefore \sum_{n=1}^{\infty} \alpha_n = \infty$$

(8) Temperature drift is smaller than the contraction

combining (6) & (7)

$$0 \leq \frac{\delta_n}{\alpha_n} \leq \frac{\Delta}{2cA} \frac{(n+2)^{\rho}}{(n+1)\left[\log(n+2)\right]^{d/2}} \to 0$$

because $\rho - 1 < 0$. Also covers $\Delta = 0$ when $\delta_n = 0$

(9) Recursion for tracking moving Gibbs law

At step n, conditional on $X_{n-1} = x$ algorithm uses exactly $P_{T_n}(x, \cdot)$

Total probability law gives $\mu_n = \mu_{n-1} P_{T_n}$

By stationarity of Π_n, contraction & triangle inequ

$$e_n = \| \mu_{n-1} P_{T_n} - \pi_n P_{T_n} \|_{TV}$$
$$\leq (1-\alpha_n) \| \mu_{n-1} - \pi_n \|_{TV}$$
$$\leq (1-\alpha_n)(e_{n-1} + \delta_n)$$
$$\leq (1-\alpha_n) e_{n-1} + \delta_n$$

(10) Convergence to global minimum in prob.

Fix any $\gamma > 0$, Part 8 gives N such that $\delta_n < \gamma \alpha_n$ for $n \geq N$

Thus, $e_n - \gamma \leq (1-\alpha_n)(e_{n-1} - \gamma)$

Taking +ve parts & iterating,

$$(e_n - \gamma)_+ \leq (e_{N-1} - \gamma)_+ \prod_{j=N}^{n} (1-\alpha_j) \leq (e_{N-1} - \gamma)_+ \exp\left(-\sum_{j=N}^{n} \alpha_j\right) \to 0$$

Part (7) proves last limit. $\therefore$ lim sup $e_n \leq \gamma$

Since $\gamma > 0$ is arbitrary and $e_n \geq 0$, we have $e_n \to 0$

finally from (8) & $T_n \to 0$ implies

$$\mathbb{P}(f(X_n) \geq f_* + \epsilon) = \mu_n(B_\epsilon) \leq e_n + \frac{V}{\nu_\epsilon} e^{-\epsilon/2T_n} \to 0$$

Consequently, $f(X_n) \xrightarrow{\mathbb{P}} f_*$ for every initial distribution supported on K. Objective value convergence.

AI Usage: ChatGPT 5o1 6.1 was used to understand the problem and help with solution suggestions checking, refining as needed for the final submission.